Nexivarin 10 mg Tablets — eCTD Submission

Module 3 — Quality

Submission Type	NDA — 505(b)(1)	**Sequence Number**	0000
Submission Date	2021-03-15	**Section**	3.2.P.2
Prepared by	PharmaCorp Inc.	**Confidentiality**	Proprietary and Confidential

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PharmaCorp Inc.
NDA 123456

Nexivarin 10 mg Tablets

Section 3.2.P.2
Drug Product — Pharmaceutical Development

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3.2.P.2 Pharmaceutical Development

3.2.P.2.1 Components of the Drug Product

The formulation development approach for Nexivarin 10 mg Film-Coated Tablets was guided by the Quality by Design (QbD) principles outlined in ICH Q8(R2). The Quality Target Product Profile (QTPP) was defined at the outset of development.

QTPP Element	Target	Justification
Dosage form	Film-coated tablet	Patient preference; marketability; manufacturing feasibility
Route of administration	Oral	Established route for antihypertensive agents
Dosage strength	10 mg	Supported by Phase II dose-response
Pharmacokinetic profile	AUC and Cmax consistent with once-daily dosing	t½ of 38 h supports accumulation to steady-state in 7 days
Drug release	Immediate release (≥80% in 30 min)	Not sustained release; full bioavailability required
Container closure	HDPE bottle, 30-count	Protects from moisture; patient-friendly
Shelf life	24 months at 15–30°C	Commercial viability requirement

3.2.P.2.2 Drug Substance

Nexivarin hydrochloride (BCS Class II) has low aqueous solubility (2.8 mg/mL at 25°C) but high intestinal permeability (Papp A→B = 18.4 × 10-6 cm/s in Caco-2 cells). Particle size control (d90 < 100 µm) was identified as a Critical Material Attribute (CMA) to ensure consistent dissolution. Micronization was explored but not required due to acceptable dissolution performance of the milled API.

3.2.P.2.3 Excipient Selection

Excipient	Selection Rationale	Compatibility
MCC PH102	Provides compressibility; inert filler-binder	Confirmed compatible by binary mixture stability (6 months, 40°C/75%RH)
Lactose Monohydrate	Co-diluent; improves flowability	No Maillard reaction observed at commercial API load (5.56% w/w)
Croscarmellose Sodium	Superdisintegrant — ensures immediate release	No incompatibility identified
Colloidal Silicon Dioxide	Improves powder flowability (glidant)	Compatible; no reactivity
Magnesium Stearate	Lubrication during tablet compression	Minimal water uptake at 1.1% w/w; no interaction with API
Opadry White	Cosmetic film coat; moisture barrier	No interaction with tablet core; moisture uptake <0.05%

3.2.P.2.4 Formulation Development

Early prototype formulations were screened using DoE (Design of Experiments) approaches. Three critical formulation variables were identified via a Plackett-Burman screening design: API particle size (d90), MCC:lactose ratio, and disintegrant concentration. A central composite design (CCD) was used to optimize the formulation. The final composition was selected based on maximizing dissolution performance (≥80% in 30 min at pH 6.8) while achieving acceptable hardness (10–14 kP), friability (<0.5%), and content uniformity (AV <5.0).

3.2.P.2.5 Manufacturing Process Development

The manufacturing process (wet granulation via high-shear granulation followed by fluid bed drying, milling, blending, and tablet compression with film coating) was selected based on the API's physical properties and required blend uniformity. Scale-up from bench (2 kg) to pilot (50 kg) to commercial scale (500 kg) was performed with no significant changes to the quality profile.